

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHONOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8384**

AY:	2025-26	Session:	Odd MSE October 2025	Date:	08-10-2025
Sem:	1	Slot:	10:00am - 11:00am	C. Code:	CS413
Exam Type:	Written	Block:	305	Seat No:	2022700014
Course Section:	1				

L_41_626_986_L_1505_41348



Name of the candidate. Dhruv Abhijeet Hogude

Candidate's UID number:

2	0	2	2	7	0	0	0	1	4
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Examination class room number:

3	0	5
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Invigilator's signature with date:

 8/10

(To be filled in by the candidate)

EXAMINATION: MSE
(For example FY MCA/FY BTECH)

SEMESTER :

PROGRAM: BE CSE - DS

I / II / III / IV / V / VI / VII / VIII

DAY AND DATE: Wednesday 7/10/25

COURSE NAME: Deep learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



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	Q.1	
	a)	
		$f(w) = \frac{1}{2} \ wx - y\ ^2$
		for gradient descent
		$\Delta w = \eta \sum_i^d (t - o) w_d$
		$w_i = w_i + \Delta w_i$
		In the gradient descent rule are applied to reduce the loss function $J(w)$ by updating the weig parameter weight while learning.
		$f'(w) = \frac{1}{2} \frac{d}{dw} \frac{1}{2} (2 \times (xw - y))$
		$f'(w) = xw - y$

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Q.1

b)

Using the expected squared errors of a predictor we can derive a bias-variance decomposition,

we can see that, ~~let~~ when the bias is higher i.e. model is trained toward specific features usually underfits the model & accuracy is too low for training set itself.

But for higher variance, model is too well trained on the training set that it captures the noise too which results in higher accuracy in training set but too fails in testing set.

When the capacity is too high, then we can see that there are higher chances of model to be overfit. As due to more capacity the model tends to learn the unwanted features & noise too which help the model to train too well with training data @ set and due to the noise, ~~the~~ unwanted features are learned due to which model fails on the test data & fails to generalize the model overall. That's why to generalize the model dimensional reduction & must be done to avoid high variance i.e. overfitting of the model.

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Q.2.
a)

Given size of input = $H \times W \times C$
with K filter size $F \times F$

So for calculation the parameters required
are

~~are~~ no. of multiplications = $F \times F \times K$

$\therefore$ The computation complexity for the
given scenario would be $F \times F \times K$

If the parameters are too large the
computation complexity would be too
high & it would be expensive & time
consuming.

Also by using ~~convolution~~ convolution we can have
the local neighbour inputs & relation & from
which we would be able to find patterns
& feature more easily & computationally fast.

If it is just a 2D convolution the computation
cost i.e. no. of multiplications will be
too high but if we consider a depthwise
separable convolution then the no. of parameters
& multiplications would decrease drastically
& will result in faster & efficient
learning.

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Q.2
b]

Undercomplete autoencoders:

The autoencoders with a ~~lower~~ smaller hidden layers is known as undercomplete autoencoders which generates compressed input ~~to~~ ~~with~~ ~~out~~ ~~the~~ features maps input pattern present in the input. Due to which learning those pattern for model is much easier & with this after decoding we can also get original input back. So in short, it is a type of feature engineering which reduce the high dimensional features. ~~to~~

Similarly, PCA is also used for dimensionality reduction as high dimensional input, ~~as~~ ~~more~~ ~~data~~ ~~is~~ ~~present~~ ~~then~~ ~~due~~ ~~high~~ ~~dimensional~~ ~~data~~ ~~processing~~ ~~will~~ ~~also~~ ~~increase~~ ~~exponential~~ ~~&~~ ~~noise~~ & unwanted patterns may get learnt, hence it is required to reduce dimensionality using PCA.

Hence we can say that linear undercomplete autoencoders trained with MSE learn a subspace equivalent to PCA.

Overcomplete autoencoders:

The autoencoders with large layer is known as overcomplete autoencoders which generate the output with higher dimensions. Due to the overcomplete autoencoders all the features get passed to the output with ~~a~~ noisy & unwanted parameters too, so it's not the correct output required for model training, hence it tends to learn the identity functions. But with regularization like denoising, sparse & compressive we can achieve the feature

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Q.3.
b]

In GRU (Gate recurrent unit), they are just two gates, update gate & reset gate.

$$\begin{aligned} \text{Update gate } u_t &= f(w_i x_i + w_h h_{t-1} + b) \\ \text{reset gate } r_t &= f(w_h x_t + w_o h_t + b) \end{aligned}$$

where, f is an activation function is most sigmoid function.

w_i is weight of input state.

x_i is input

w_h is weight of hidden state

h_{t-1} previous state

b is a bias

w_o is weight of output state

States :

$$h_t = f(r_t \odot h_{t-1} + u_t \odot x_i + c)$$

$$y_t = (1 - r_t) \odot h_t + c$$

Considering these states of a GRU, we can see that update^{reset} gate works as ~~forget~~ forget gate & input gate & output gate. The update gate tries to forget the unwanted parameter with adding the new input & reset gates works as output gate ~~for~~ given. In this way we can see that update rule is convex combination of previous hidden state (h_{t-1}) & candidate hidden state ($\hat{h}_t$) which is

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Q As we can see that GRU has 2 gates & 2 states while ~~LSTM~~ LSTM has 3 gates forget gate, input gate & output gate with respective states.

Due to the fewer gates i.e. fewer parameters in GRU the computation is faster than the LSTM but accuracy for LSTM could be higher than the GRU.

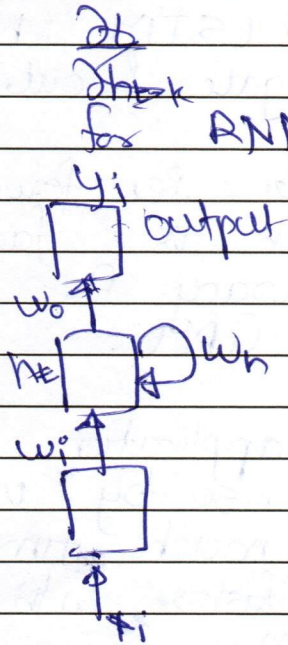
For ~~image~~ video captioning application, the video can be much ~~to~~ larger, so by us LSTM for this would take ~~to~~ much time so GRU would be result faster with comparable accuracy with the LSTM.

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Q.3

g)



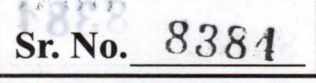
$$h_t = f(w_i \times x_i + w_h \times h_{t-1} + b)$$

$$y_i = f(w_o \times h_t + b)$$

where

 h_t = hidden state x_i = input y_i = output w_i = input weight w_o = output weight w_h = hidden weight

$$\frac{\partial L}{\partial h_t} = \frac{\partial L}{\partial f(w_i \times x_i + w_h \times h_{t-1} + b)}$$

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